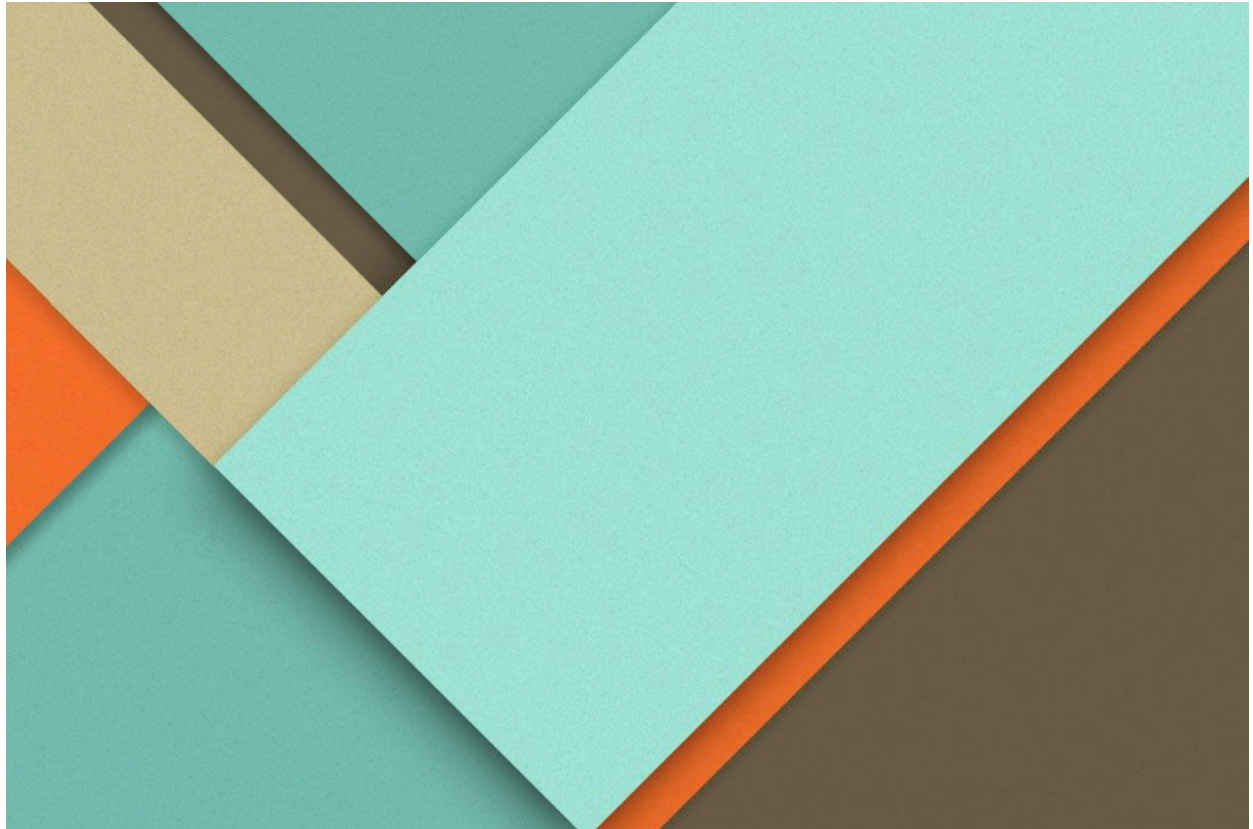




“The Guardian”: A Modular AI for Proactive Human-Layer Cybersecurity on Social Media

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Final Year Research Project
SLIIT, FoC, Cyber Security



We aim to build "**The Guardian**" – a modular browser extension that uses AI to alert users in real-time about human-layer threats on social media. Each group member is researching and developing a specialized NLP module. We will evaluate accuracy, performance, and usability, and contribute practical, ethical cybersecurity solutions for everyday users.

1. Introduction to the Problem (With Research Context)

Cybersecurity threats have evolved beyond viruses and technical exploits. Social media platforms have become hotspots for **human-layer threats**: attacks that exploit **psychology, emotion, and human behavior**. Like:

- Hate speech and harassment
- Misinformation and disinformation
- Social engineering and online scams/phishing
- Unintentional leaks of Personally Identifiable Information (PII)

While platforms like Facebook and Twitter do have **moderation systems**, most are **reactive**, triggered **after** content is reported.

Research Gap:

Existing tools are often:

- Focused on **content moderation**, not **user guidance**
- Limited to **single threat types**
- Lacking in **real-time, browser-based, modular solutions**

While most tools are reactive, platform-dependent, and limited to one threat, "The Guardian" is: **Modular** (detects multiple threat types), **Real-time** (works as you browse), **User-guiding** (not censoring), **Platform-agnostic** (browser extension, not API-dependent)

Real-World Systems vs. Our Proposal

Existing System	What It Does	Limitations
Twitter/X Community Notes	Crowdsourced fact-checks on tweets	Only works on viral tweets; not real-time
Facebook AI Moderation	Detects hate speech & misinformation	Works server-side, no user control
Norton Safe Web / Trend Micro	Detects malicious URLs	Doesn't analyze human language
NewsGuard	Rates website credibility	Doesn't work on social media post content
AI Detector for Hate Speech (Davidson et al., 2017)	Classifies text into hate speech/offensive/neutral	High false positive rate on sarcasm & coded language

Why this is novel

Feature	Novelty	Literature Support
Modular detection	Each module is independent and specialized	Rare in current systems
Client-side browser-based	Real-time user protection, not censorship	Ghafir et al. (2018), Al-Rimy et al. (2018)
PII + OpSec modeling	Few studies address personal info + behavioral risks	Custom NER + rule-based blend
Gentle warning UI	Avoids disruption or censorship	Inspired by HateXplain's explainable model focus
Multi-model comparison	You'll evaluate BERT vs DistilBERT per module	Valid comparative research

2. Research Aim & Objectives

To design, implement, and evaluate "**The Guardian**", a modular, AI-powered browser extension that proactively alerts users to human-layer threats while browsing social media, in real time and in a non-intrusive manner.

Our Sub-Objectives:

Each group member will focus on building and researching one threat-detection module using NLP and deep learning.

Member	Focus Area
Member 1	PII & OpSec Risk Detection
Member 2	Hate Speech Detection
Member 3	Misinformation Detection
Member 4	Scam/Phishing Detection

3. How the System Works (Architecture Overview)

System Flow:

1. User is browsing Facebook or Twitter.
2. Our **Chrome/Firefox extension** extracts visible text (not API scraping).
3. The extension sends the text to our **FastAPI backend**.
4. The backend calls each of the **4 AI modules** for analysis:
 - Hate speech
 - Misinformation
 - Scam/Phishing
 - PII & OpSec risks
5. Each module analyzes the text and returns a JSON result.
6. The browser extension displays **a subtle warning icon** with a tooltip if a threat is detected.

(The Gantt Chart and the System Designs will be shared separately)

Example Use Case (With Research-Linked Justification)

User writes: "I'm going to India next week! Call me at 0771234567 if anything happens."

Our system: Uses **NER (Named Entity Recognition)** to detect phone number and location → flagged as **PII**

- Applies **OpSec rule-based patterns** ("next week" + location) → flagged as **risk of travel exposure**

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- Shows a tooltip: “⚠️ You may be sharing personal details that could be used in scams or identity theft.”

Linked to:

- **Ghafir et al.** → Emphasizes OpSec risks
- **spaCy / Hugging Face models** → Enables NER for custom entities
- **Zellers et al.** → Shows importance of context-aware understanding

4. Research Breakdown by Team Member

Member	Research Questions
Member 1 (PII/OpSec)	How accurate are existing NER models at detecting PII in social media posts? Can we improve this with custom rules?
Member 2 (Hate Speech)	Can NLP detect implicit and coded hate speech (“dog whistles”)? What are the fairness issues?
Member 3 (Misinformation)	Can a model detect manipulative language without a fact-checking database?
Member 4 (Scams)	What linguistic patterns are most common in social media scams? Can we detect grooming or emotional manipulation?

Each member will:

- Compare models (e.g., BERT vs DistilBERT)
- Fine-tune or evaluate using specific datasets
- Test detection performance on real-world examples

Data/ Datasets:

We will use **public datasets** and supplement with small custom samples if needed.

- Hate Speech: Davidson dataset, HateXplain
- Misinformation: LIAR dataset, COVID disinfo
- Scams: Social media scam datasets from Kaggle or create manually
- PII: spaCy NER, create test data with Sri Lankan context



Module	Dataset	Source
Hate Speech	Davidson et al., HateXplain	Academic datasets
Misinformation	LIAR, CoAID	Hugging Face, Kaggle
PII / OpSec	spaCy NER, custom Sri Lankan samples	Create + augment
Scams	Kaggle Scam Messages, Twitter scam patterns	Public datasets

Models:

Use **pretrained Transformer models** from Hugging Face (BERT, DistilBERT)

For PII: Compare spaCy's NER with BERT-based NER

Fine-tune where appropriate

Testing:

Each module will be evaluated using:

- Precision, recall, accuracy
- False positive and negative rates
- User interface feedback (optional)

Expected Results & Evaluation Criteria:

Goal	How We Measure It
Detect threats accurately	Model evaluation metrics (F1-score, etc.)



Work in real-time	Response time of browser extension
Be user-friendly	UI should not interrupt browsing
Preserve privacy	No data storage, only local processing

7. Technologies Used

Area	Tools
AI/NLP	Python, Hugging Face Transformers, spaCy
Backend	FastAPI
Frontend	JavaScript, HTML/CSS (for browser extension)
Collaboration	GitHub
Testing	Postman, Jupyter Notebooks

8. Why It's a Valid Research Project

This is not just software development — it's **applied research** because:

- We're **exploring AI's effectiveness in a new problem domain** (real-time, human-layer social media security).
- We're conducting **module-specific evaluations** on real-world text threats.
- We're proposing a **modular, scalable, privacy-focused system**, which hasn't been practically tested this way.

The novelty lies in:

- The integration of multiple AI models for **live social media text**
- Focusing on **user guidance, not censorship**
- Bringing **cybersecurity + AI + UX** together in a browser tool

Proposed Plan Feasibility:

Month	Activity
Month 1-2	Learn ML, NLP, Transformers
Month 2-3	Dataset collection & initial model testing
Month 3-5	Build each module individually
Month 5-6	Backend + browser extension integration
Month 6-7	System testing and fine-tuning
Month 8	Report writing, user feedback, presentation prep

9. Literature Review – Key Studies

1. Davidson et al. (2017) – *Automated Hate Speech Detection*

- Dataset of ~25k tweets labeled for hate/offensive language
- Highlighted the challenge of **false positives** due to **slang, sarcasm, and context**

2. Zellers et al. (2019) – *HellaSwag*

- Studied how models can reason about language in context
- Motivated use of **Transformer-based models** like BERT for NLP tasks

3. Ghafir et al. (2018) – *Human Factor in Cyber Threats*

- Emphasized that **human psychology is the weakest link** in cybersecurity
- Highlighted need for real-time, **user-centric solutions** (your system fits this well)

4. HateXplain (Mathew et al., 2020) – Multitask hate speech dataset with human rationales

- Used **explanations** to improve hate speech classifier trustworthiness
- Supports your design goal of using **explainable tooltips**, not just warnings

5. Al-Rimy et al. (2018) – *Survey on Ransomware and Human Threats*

- Argued that traditional technical defenses **ignore social vectors**
- Motivates your project's **focus on language-based, human-side attacks**